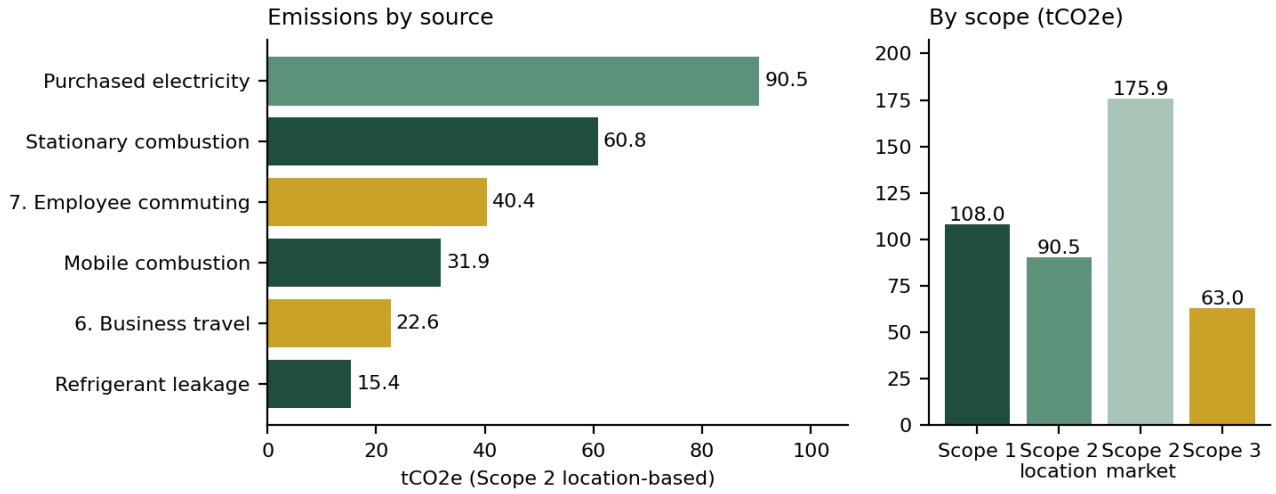
Corporate Carbon Footprint Assessment

GHG Protocol Corporate Standard | reporting year 2025 | prepared by Muhammed Najeeb Choondathody

Executive summary. This report assesses the greenhouse gas emissions of Nordovest Precision Parts S.r.l., a fictional 200-person manufacturer with two sites in Italy. All activity data are invented to illustrate a complete assessment: inventory, data controls, Scope 3 screening, reduction roadmap and ESRS E1 mapping. Emission factors come from published sources, listed in section 3 with their verification status.

Scope 1	Scope 2 (location)	Scope 2 (market)	Scope 3 (partial)
108.0 tCO_2e	**90.5** tCO_2e	**175.9** tCO_2e	**63.0** tCO_2e

Total: 261.5 tCO_2e location-based and 347.0 tCO_2e market-based. The two Scope 2 methods are shown side by side and never added together.



Key messages

- Natural gas for heating is the largest direct source (60.8 tCO_2e, 23% of the location-based total); purchased electricity is the largest single source overall (90.5 tCO_2e).
- The market-based method raises Scope 2 from 90.5 to 175.9 tCO_2e, because the residual mix is more carbon-intensive than the grid average.
- Scope 3 is partial: 2 of 15 categories are quantified, and purchased goods and services, the likely largest category, is still missing. Read the Scope 3 figure as a floor.
- An illustrative roadmap cuts the market-based total by 60% but the location-based total by only 12%. Most of the market-based gain comes from renewable electricity procurement, which does not change grid emissions.
- Three of seven factors are still SECONDARY (taken from a page that quotes the official dataset) and must be confirmed before any formal use.

1. Scope and boundary

Entity: Nordovest Precision Parts S.r.l. (fictional). **Period:** 1 January to 31 December 2025, also the base year. **Consolidation:** operational control. **Gases:** CO_2, CH_4, N_2O and HFCs, converted with IPCC AR5 100-year GWPs (CH_4 = 28, N_2O = 265, the values ISPRA uses).

Site	Country	Activity	Included
HQ Milan	Italy	Offices	Yes
Plant 1	Italy	Manufacturing	Yes

- Scope 3 categories 1, 3, 4 and 5 are relevant but not quantified in this version (see Scope3_Screening).
- Categories 8, 11, 13, 14 and 15 are judged not relevant for this business; rationale in Scope3_Screening.
- Fuel factors cover CO2 only; CH4 and N2O from combustion are not included. No process emissions or biogenic CO2 sources are assumed.
- Both sites are within the operational-control boundary; no joint ventures or leased-out assets.

2. Method

Every line follows one calculation: **activity data x emission factor = emissions**. Factors are stored in kg CO_2e per unit and results are shown in tonnes.

- **Scope 1:** diesel for fleet and generator, natural gas for boilers, refrigerant leakage (R-410A).
- **Scope 2:** purchased electricity, calculated twice with the same kWh: *location-based* (Italian grid consumption factor) and *market-based* (national residual mix).
- **Scope 3:** all 15 categories screened and the decision recorded; categories 6 (business travel) and 7 (employee commuting) quantified.
- **Traceability:** each activity row records its data owner, source document and a quality score (1 = metered or invoiced, 5 = rough proxy). Each factor records publisher, dataset, year and status.
- **Controls:** 12 automated checks cover missing or invalid factors, unit mismatches, missing sources, Scope 2 reconciliation between methods, and totals across scopes and categories.

3. Activity data and emission factors

Row	Scope	Source	Quantity	Factor ID	Data source	DQ
A-001	1	Diesel for fleet and generator (from delivery notes)	10 tonne	EF-001	Fuel delivery notes (Fleet manager)	2
A-002	1	Natural gas for boilers	30,000 m3	EF-002	Utility invoices (Facilities)	1
A-003	1	R410A top-up in chillers	8 kg	EF-003	Maintenance log (Facilities)	2
A-004	2 (loc.)	Grid electricity	450,000 kWh	EF-004	Utility invoices (Facilities)	1
A-005	2 (mar.)	Grid electricity	450,000 kWh	EF-005	Utility invoices (Facilities)	1
A-006	3	Air travel, short-haul economy	180,000 pkm	EF-006	Travel agency report (HR)	2
A-007	3	Commuting by car (survey-based)	250,000 km	EF-007	Employee survey (HR), extrapolated	4

Emission factor register

VERIFIED: the value was read in the primary publication. **SECONDARY**: taken from a secondary page that quotes the official dataset; to be confirmed against the primary file before formal use.

ID	Factor	Value	Source (year)	Status
EF-001	Diesel (gasolio), stationary and mobile combustion, CO2 only	3186 kg CO_2/tonne	ISPRA Rapporti 430/2026, Table 1.4 (IPCC 2006 default, 3.186 tCO2/t) (2006)	**VERIFIED**
EF-002	Natural gas, stationary combustion, CO2 only	2.026 kg CO_2/m3	ISPRA Rapporti 430/2026, Table 1.4 (2.026 tCO2 per 1000 Sm3) (2024)	**VERIFIED**
EF-003	Refrigerant R-410A leakage (GWP-weighted)	1924 kg CO_2e/kg	IPCC AR5 100-year GWP, as stated in Arkema Forane 410A datasheet (2013)	**SECONDARY**
EF-004	Italian grid electricity, location-based (consumption mix, GHG)	0.20101 kg CO_2e/kWh	ISPRA Rapporti 430/2026, Table 1.18 (201.01 g CO2e/kWh) (2025)	**VERIFIED**
EF-005	Italian residual mix, market-based proxy (CO2)	0.3909 kg CO_2/kWh	ISPRA Rapporti 430/2026, section 1.3 (residual mix 390.9 g CO2/kWh) (2025)	**VERIFIED**
EF-006	Air travel, short-haul, economy, with radiative forcing	0.12576 kg CO_2e/pkm	DESNZ/DEFRA 2026, sheet 'Business travel-air', cell E25 (via greencalculus.com and carbonpass.co) (2026)	**SECONDARY**
EF-007	Passenger car, average petrol, engine size unknown	0.16152 kg CO_2e/km	DESNZ/DEFRA 2026, sheet 'Business travel-land' (via greencalculus.com) (2026)	**SECONDARY**

Known limitations of the factors

- Gas and diesel factors cover CO_2 only, which slightly understates Scope 1.
- The 2025 electricity factors are preliminary and the gas factor is the latest final year (2024); a real inventory should align the years.
- The market-based factor is the Italian residual mix from the same ISPRA report (CO_2 only, production basis), not the AIB European Residual Mix or a supplier-specific factor.
- Air and car factors come from the UK government set (DESNZ 2026) and approximate Italian operations.

4. Results

Row	Scope	Source	Quantity	Factor	tCO_2e	Share*
A-001	1	Diesel for fleet and generator (from delivery notes)	10 tonne	3186 kg/tonne	31.9	12.2%
A-002	1	Natural gas for boilers	30,000 m3	2.026 kg/m3	60.8	23.2%
A-003	1	R410A top-up in chillers	8 kg	1924 kg/kg	15.4	5.9%
A-004	2 (loc.)	Grid electricity	450,000 kWh	0.20101 kg/kWh	90.5	34.6%
A-006	3	Air travel, short-haul economy	180,000 pkm	0.12576 kg/pkm	22.6	8.7%
A-007	3	Commuting by car (survey-based)	250,000 km	0.16152 kg/km	40.4	15.4%
		Total, location-based			**261.5**	**100%**

* Share of the location-based total. The market-based Scope 2 row (same 450,000 kWh) gives a market-based total of 347.0 tCO_2e.
Emissions-weighted data quality score: 1.4 (1 = best, 5 = weakest).

5. Scope 3 screening

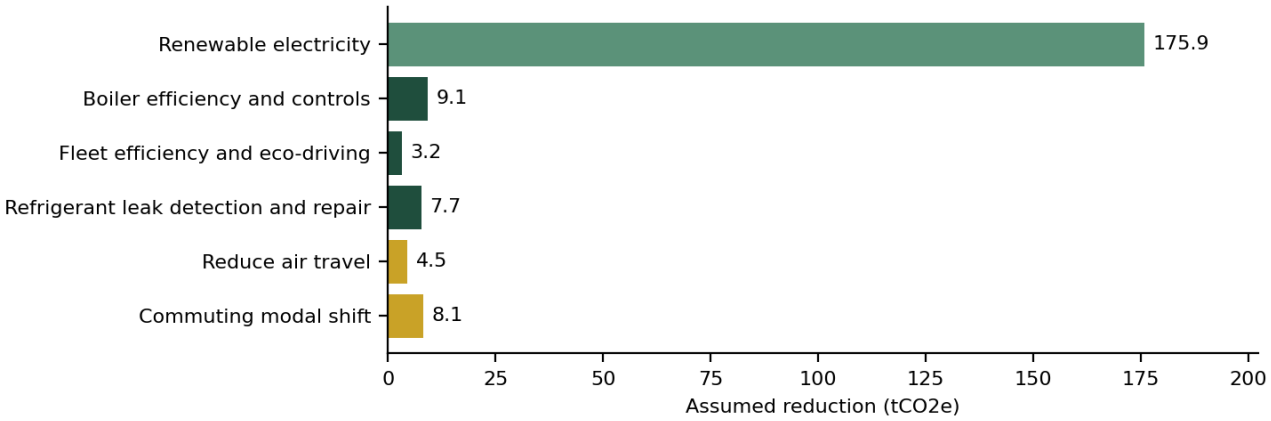
All 15 categories were reviewed. The result shows which parts of the value chain are measured and which are not.

Category	Relevant	Quantified	Rationale
1. Purchased goods and services	Yes	No	Main category for a manufacturer: metals, components and packaging. Not quantified in this version; needs purchase data and spend-based or supplier factors.
2. Capital goods	Partly	No	Machinery purchases are lumpy from year to year. Not quantified.
3. Fuel- and energy-related activities	Yes	No	Upstream emissions of fuels and electricity, and grid losses. Not quantified; factors exist in the UK government set (well-to-tank) but were not retrieved.
4. Upstream transportation and distribution	Yes	No	Inbound freight of materials. Not quantified; needs tonne-km from logistics providers.
5. Waste generated in operations	Yes	No	Production waste and scrap handled by contractors. Not quantified; waste records exist by stream.
6. Business travel	Yes	Yes	Quantified: short-haul economy flights from the travel agency report (passenger-km).
7. Employee commuting	Yes	Yes	Quantified: car commuting estimated from a survey of 60 of 200 employees and extrapolated.
8. Upstream leased assets	No	No	No leased assets beyond the offices already inside the operational-control boundary.
9. Downstream transportation and distribution	Partly	No	Customers usually arrange delivery; small share paid by the company. Not quantified.
10. Processing of sold products	No	No	Parts are shipped as components; downstream processing is not known and is excluded.
11. Use of sold products	No	No	Components have no direct energy use in operation; use-phase is attributed to the customer's product.
12. End-of-life treatment of sold products	Partly	No	Metal parts are largely recycled at end of life; small expected effect. Not quantified.
13. Downstream leased assets	No	No	The company does not lease assets to others.
14. Franchises	No	No	No franchise operations.
15. Investments	No	No	No equity or debt investments in the reporting scope.

Four relevant categories (1, 3, 4, 5) are not yet quantified, so the Scope 3 total understates the value chain.

6. Hotspots and reduction roadmap

The percentages below are illustrative assumptions, set in the workbook so they can be changed. They are not company commitments.



Measure	Scope	Baseline	Assumed cut	Reduction (tCO_2e)
Renewable electricity (guarantees of origin or PPA)	2	175.9	100%	175.9
Boiler efficiency and controls	1	60.8	15%	9.1
Fleet efficiency and eco-driving	1	31.9	10%	3.2
Refrigerant leak detection and repair	1	15.4	50%	7.7
Reduce air travel	3	22.6	20%	4.5
Commuting modal shift	3	40.4	20%	8.1

- **Location-based total:** 261.5 to 228.9 tCO_2e (12% lower). This is the physical reduction.
- **Market-based total:** 347.0 to 138.4 tCO_2e (60% lower), mostly from renewable electricity procurement.
- **Illustrative target check:** a 30% cut in market-based Scope 1 + 2 by 2030 means 198.8 tCO_2e; the measures reach 88.0 tCO_2e (69% cut). On a location-based basis, the same measures cut Scope 1 + 2 by 10%, so the target would not be met without the renewable contract.
- **Priorities:** improve gas use first (the largest direct source), secure renewable supply for the market-based figure, fix leaks (small mass, high GWP: 15.4 tCO_2e), and improve commuting data before acting on it (40.4 tCO_2e, quality score 4).

7. ESG reporting layer

Materiality screening. The 10 ESRS topical standards are scored 1 to 5 for impact and for financial relevance; a topic is material if either score reaches the threshold of 3. The scores are the author's illustrative judgement, not the result of a stakeholder process.

ESRS topic	Ref	Impact	Financial	Result	Rationale
Climate change	E1	5	4	Material	Energy- and gas-intensive production; exposure to carbon costs and customer requests.
Pollution	E2	3	2	Material	Air emissions and lubricants from machining; regulated through permits.
Water and marine resources	E3	2	2	Not material	Low process water use; no high-stress site assumed.
Biodiversity and ecosystems	E4	1	1	Not material	Urban industrial sites; no protected areas nearby (assumed).
Resource use and circular economy	E5	4	3	Material	Metal inputs and scrap: material efficiency drives cost and supply risk.
Own workforce	S1	4	3	Material	Manufacturing workforce: health and safety, skills and retention.
Workers in the value chain	S2	3	3	Material	Metals and component suppliers; due-diligence expectations from customers.
Affected communities	S3	2	1	Not material	Plant in an industrial area; limited noise and traffic effects.
Consumers and end-users	S4	1	1	Not material	B2B components; no direct consumers.
Business conduct	G1	3	3	Material	Supplier and customer ethics; anti-corruption policies required by large customers.

ESRS E1 mapping. ESRS and CSRD are being revised under the EU Omnibus simplification, so this mapping should be checked against the current text.

Disclosure	Title	Workbook source	Status
E1-1	Transition plan for climate change mitigation	Not in this workbook. The baseline in Results is the starting point.	In progress
E1-2	Policies related to climate change mitigation and adaptation	Not in this workbook.	Not started
E1-3	Actions and resources in relation to climate change policies	Not in this workbook. Largest sources in Results guide where to act.	In progress
E1-4	Targets related to climate change mitigation and adaptation	Base year (Boundary) and baseline emissions (Results).	Data available
E1-5	Energy consumption and mix	Activity_Data fuel and electricity rows. Renewable share and energy conversion not yet included.	In progress
E1-6	Gross Scopes 1, 2, 3 and total GHG emissions	Results rows 7-12 and the category table.	Data available
E1-7	GHG removals and carbon credits	Not in this workbook.	Not started
E1-8	Internal carbon pricing	Not in this workbook.	Not started
E1-9	Anticipated financial effects from material physical and transition risks	Not in this workbook.	Not started

8. Data governance and assumptions

Data item	Scope	Owner	Control
Diesel purchases	1	Fleet manager	Delivery notes reconciled to supplier invoices
Natural gas consumption	1	Facilities manager	Invoice totals reconciled to meter readings; change above 20% month on month investigated
Refrigerant top-ups	1	Maintenance lead	Quantity checked against F-gas register
Electricity consumption	2	Facilities manager	Invoice kWh reconciled to sub-meter totals
Business flights	3	HR / travel desk	Passenger-km recomputed for a sample of 10 trips
Employee commuting	3	HR	Response rate and extrapolation documented in assumptions log
Emission factor library	1-3	ESG analyst	Every factor has source, year and status; SECONDARY factors flagged
Inventory model	1-3	ESG analyst	12 automated checks on the Controls sheet; second-person review before release

ID	Assumption	Impact
AS-001	Commuting distance extrapolated from a survey of 60 of 200 employees to full headcount.	Medium
AS-002	Diesel for fleet and generator combined into one quantity from delivery notes.	Low
AS-003	Market-based Scope 2 uses the ISPRA national residual mix (CO2 only, production basis) as a proxy.	Medium
AS-004	Factor years differ: gas 2024 (latest final), electricity 2025 (preliminary).	Low
AS-005	UK government (DESNZ) travel factors applied to Italian travel and commuting.	Medium
AS-006	Scope 3 limited to categories 6 and 7; categories 1, 3, 4 and 5 are relevant but not quantified.	High
AS-007	Reduction percentages and the 2030 target in Reduction_Scenarios are illustrative assumptions.	Medium

9. Limitations

- The company and its activity data are fictional: the results demonstrate the method, not a real footprint.
- Scope 3 is partial (2 of 15 categories); purchased goods and services is the main gap.
- Three factors are SECONDARY and must be confirmed in the official DESNZ 2026 spreadsheet (and the R-410A GWP in IPCC AR5) before formal use.
- Fuel factors are CO_2 only, the market-based factor is a national proxy, and factor years are mixed.
- Reduction percentages and the 2030 target are assumptions; no cost, feasibility or timing analysis is included.
- There is no uncertainty analysis; data quality scores are the only reliability indicator.

10. Next steps

- Confirm the three SECONDARY factors in the primary sources and set them to VERIFIED.
- Add a spend-based estimate for purchased goods and services, flagged as lower quality, then upstream energy (category 3) and inbound freight (category 4).
- Replace the market-based proxy with supplier-specific or AIB data, and align factor years.
- Add uncertainty ranges, a base-year recalculation policy and an independent review of the inventory.
- Replace the illustrative roadmap with costed measures agreed with the site managers.

Sources

ISPRA, *Settore elettrico: emissioni di CO2 e altri impatti, Edizione 2026*, Rapporti 430/2026 (Tables 1.4 and 1.18, section 1.3). UK Government GHG Conversion Factors for Company Reporting 2026 (DESNZ). IPCC AR5 100-year GWPs. Arkema, Forane 410A datasheet. GHG Protocol Corporate Accounting and Reporting Standard, Scope 2 Guidance and Scope 3 Standard.